All phase/repair transfer cells; 32 paired initials per scenario
* physical error bound exceeded; † baseline failure; × incomplete controller; no transfer certificate

Phase 50 ms / 1802

	Nominal	Gain/drag	Mild lag	Strong lag
Ordinary	0.56	0.06	0.05	0.02*
Dense	0.73	0.97	0.30	0.51*†×
Slow yield	-0.04	0.01	0.27	0.01*†×
Unequal speed	0.07	0.10	0.04	0.17*
Tight gap	0.85	0.39	0.79	9.69*†

Phase 50 ms / 1812

	Nominal	Gain/drag	Mild lag	Strong lag
Ordinary	0.55	0.05	0.06	0.02*
Dense	0.26	0.79	0.18	3.73*†×
Slow yield	-0.04	0.21	0.01	0.24*†×
Unequal speed	0.05	0.10	0.06	0.10*
Tight gap	0.96	0.53	0.78	5.24*†×

Phase 50 ms / 1822

	Nominal	Gain/drag	Mild lag	Strong lag
Ordinary	0.53	0.04	0.03	0.12*
Dense	0.42	0.59	0.22	0.57*†×
Slow yield	0.05	0.24	0.01	0.43*†×
Unequal speed	0.03	0.08	0.01	0.09*
Tight gap	1.13	0.77	0.95	5.52*†×

Phase 100 ms / 1802

	Nominal	Gain/drag	Mild lag	Strong lag
Ordinary	0.58	0.07	0.06	0.03*
Dense	0.73	0.97	0.31	0.51*†×
Slow yield	-0.04	0.01	0.28	0.01*†×
Unequal speed	0.07	0.10	0.07	0.23*
Tight gap	0.91	0.58	0.79	9.69*†

Phase 100 ms / 1812

	Nominal	Gain/drag	Mild lag	Strong lag
Ordinary	0.56	0.07	0.06	0.15*
Dense	0.34	0.80	0.19	3.66*†×
Slow yield	-0.04	0.21	0.01	0.24*†×
Unequal speed	0.05	0.16	0.08	0.24*
Tight gap	0.96	0.53	0.79	5.18*†×

Phase 100 ms / 1822

	Nominal	Gain/drag	Mild lag	Strong lag
Ordinary	0.55	0.04	0.04	0.25*
Dense	0.43	0.61	0.23	0.50*†×
Slow yield	0.05	0.24	0.01	0.43*†×
Unequal speed	0.03	0.14	0.02	0.22*
Tight gap	1.20	0.78	0.95	5.52*†×

Action repair / 1802

	Nominal	Gain/drag	Mild lag	Strong lag
Ordinary	0.58	0.07	0.07	0.15*
Dense	0.73	0.98	0.31	4.46*†×
Slow yield	-0.04	0.02	0.33	0.00*†×
Unequal speed	0.06	0.11	0.07	0.29*
Tight gap	1.10	0.85	0.87	9.99*†

Action repair / 1812

	Nominal	Gain/drag	Mild lag	Strong lag
Ordinary	0.56	0.12	0.07	0.26*
Dense	0.36	0.80	0.21	3.68*†×
Slow yield	0.02	0.23	0.18	0.35*†×
Unequal speed	0.05	0.17	0.09	0.30*
Tight gap	1.10	0.86	0.85	5.38*†×

Action repair / 1822

	Nominal	Gain/drag	Mild lag	Strong lag
Ordinary	0.55	0.10	0.04	0.25*
Dense	0.43	0.61	0.23	0.81*†×
Slow yield	0.06	0.27	0.06	0.51*†×
Unequal speed	0.04	0.15	0.03	0.35*
Tight gap	1.26	1.10	1.03	5.66*†×

Mean team-cost improvement over matched baseline (%)
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5